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Question

Consider a discrete-time LTI system with input $x[n] = \delta[n] + \delta[n - 1]$ and impulse response $h[n] = \delta[n] - \delta[n - 1]$.

The output of the system will be given by:

- (a) $\delta[n] - \delta[n - 2]$ (c) $\delta[n - 1] + \delta[n - 2]$
 (b) $\delta[n] - \delta[n - 1]$ (d) $\delta[n] + \delta[n - 1] + \delta[n - 2]$

Solution

For an LTI system, the output is obtained by convolution:

$$y[n] = x[n] * h[n]. \quad (1)$$

Given input and impulse response:

$$x[n] = \delta[n] + \delta[n - 1], \quad (2)$$

$$h[n] = \delta[n] - \delta[n - 1]. \quad (3)$$

Representing them as vectors:

$$\mathbf{x} = \begin{pmatrix} x[0] \\ x[1] \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{h} = \begin{pmatrix} h[0] \\ h[1] \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (4)$$

Determine output length:

$$N_y = N_x + N_h - 1 = 2 + 2 - 1 = 3. \quad (5)$$

Form the convolution (Toeplitz) matrix using $\mathbf{h}$:

$$\mathbf{H} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix}. \quad (6)$$

Compute the output vector:

$$\begin{aligned} \mathbf{y} &= \mathbf{H}\mathbf{x} \\ &= \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1(1) + 0(1) \\ -1(1) + 1(1) \\ 0(1) - 1(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \end{aligned} \quad (7)$$

Thus, the output samples are:

$$y[0] = 1, \quad y[1] = 0, \quad y[2] = -1. \quad (8)$$

Hence, the output signal is:

$$y[n] = \delta[n] - \delta[n - 2]. \quad (9)$$

Final answer:

$$\boxed{y[n] = \delta[n] - \delta[n - 2] \quad (\text{Option (a)})} \quad (10)$$